

Experts Problem and Online Convex Optimization

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References

- Lecture 1 and 2 of [Haipeng's course](#)

Learning from Experts (The Experts Problem)

Suppose there are N experts (actions)

For $t = 1, 2, \dots, T$:

Learner generates a distribution $p_t \in \Delta_N$

Adversary decides the loss of every expert $\ell_t = (\ell_t(1), \dots, \ell_t(N)) \in \mathbb{R}^N$

Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle = \mathbb{E}_{i \sim p_t} [\ell_t(i)]$

A special case of Online Convex Optimization with

$\mathcal{X} = \Delta_N$ and $f_t(p) = \langle p, \ell_t \rangle$

$$\text{Regret} := \sum_{t=1}^T \langle p_t, \ell_t \rangle - \min_{i \in [N]} \sum_{t=1}^T \ell_t(i)$$

Follow the Leader (FTL)

Suppose there are N experts (actions)

For $t = 1, 2, \dots, T$:

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Adversary reveals ℓ_t to the learner

Learner suffers loss $\langle p_t, \ell_t \rangle$

Define
$$L_t(i) := \sum_{s=1}^t \ell_s(i)$$

“Follow the leader” strategy: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$ and let $p_t = e_{i_t}$

Follow the Leader (FTL)

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

$$\text{Regret} = \underbrace{\sum_{t=1}^T \langle p_t, \ell_t \rangle}_{\geq T-1} - \underbrace{\min_{i \in [N]} \sum_{t=1}^T \ell_t(i)}_{\approx \frac{T}{2}} \approx (T-1) - \frac{T}{2} \approx \underline{\Theta\left(\frac{T}{2}\right)} \neq o(T)$$

Softmin

Replacing “min” by “softmin”:

$$p_t(i) = \frac{w_t(i)}{\sum_{j=1}^N w_t(j)} \quad \text{where} \quad w_t(i) = e^{-\eta L_{t-1}(i)} \quad \textbf{(Exponential weights)}$$

“Follow-the-leader” (hardmin) is a special case when $\eta = \infty$

Regret Guarantee for Exponential Weights

Theorem 1.

Assume that $\eta \ell_t(i) \geq -1$ for all t, i . Then Exponential Weight ensures

$$\text{Regret} \leq \frac{\log N}{\eta} + \eta \sum_{t=1}^T \sum_{i=1}^N p_t(i) \ell_t(i)^2$$

Assume $0 \leq \ell_t(i) \leq 1$

$$\text{Regret} \leq \frac{\log N}{\eta} + \eta T = 2\sqrt{T \log N} \leq o(T)$$

$$\eta = \sqrt{\frac{\log N}{T}}$$

$$\log = \ln$$

Proof of Theorem 1

Recall $L_t(i) = \sum_{s=1}^t l_s(i)$

and $P_t(i) = \frac{w_t(i)}{W_t}$ where $w_t(i) = \exp(-\eta L_{t-1}(i))$, $W_t = \sum_{j=1}^N w_t(j)$

$$\begin{aligned} \ln \frac{W_{t+1}}{W_t} &= \ln \frac{\sum_{i=1}^N w_{t+1}(i)}{W_t} = \ln \frac{\sum_{i=1}^N \exp(-\eta L_t(i))}{W_t} \\ &= \ln \frac{\sum_{i=1}^N \underbrace{\exp(-\eta L_{t-1}(i))}_{w_t(i)} \exp(-\eta l_t(i))}{W_t} \\ &= \ln \frac{\sum_{i=1}^N w_t(i) \exp(-\eta l_t(i))}{W_t} = \ln \left(\sum_{i=1}^N P_t(i) \exp(-\eta l_t(i)) \right) \end{aligned}$$

$$(L_t(i) = L_{t-1}(i) + l_t(i))$$

$$\ln \frac{W_{t+1}}{W_t} = \ln \left(\sum_{i=1}^N P_t(i) \exp(-\eta l_t(i)) \right)$$

$$\leq \ln \left(\sum_{i=1}^N P_t(i) (1 - \eta l_t(i) + \eta^2 l_t(i)^2) \right)$$

$$= \ln \left(\sum_{i=1}^N P_t(i) - \eta \sum_{i=1}^N P_t(i) l_t(i) + \eta^2 \sum_{i=1}^N P_t(i) l_t(i)^2 \right)$$

$$= \ln \left(1 - \eta \langle P_t, l_t \rangle + \eta^2 \sum_{i=1}^N P_t(i) l_t(i)^2 \right)$$

$$\leq -\eta \langle P_t, l_t \rangle + \eta^2 \sum_{i=1}^N P_t(i) l_t(i)^2$$

For real number $z \leq 1$

$$\exp(z) \leq 1 + z + z^2$$

$$z = -\eta l_t(i)$$

$$\ln(1+z) \leq z$$

Summing over $t=1, 2, \dots, T$

$$\ln \frac{W_{T+1}}{W_1} \leq -\eta \sum_{t=1}^T \langle P_t, l_t \rangle + \eta^2 \sum_{t=1}^T \sum_{i=1}^N P_t(i) l_t(i)^2$$

$$\boxed{\ln \frac{W_{T+1}}{W_1}} \leq -\eta \sum_{t=1}^T \langle p_t, l_t \rangle + \eta^2 \sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$

$$i^* = \underset{i}{\operatorname{argmin}} L_T(i)$$

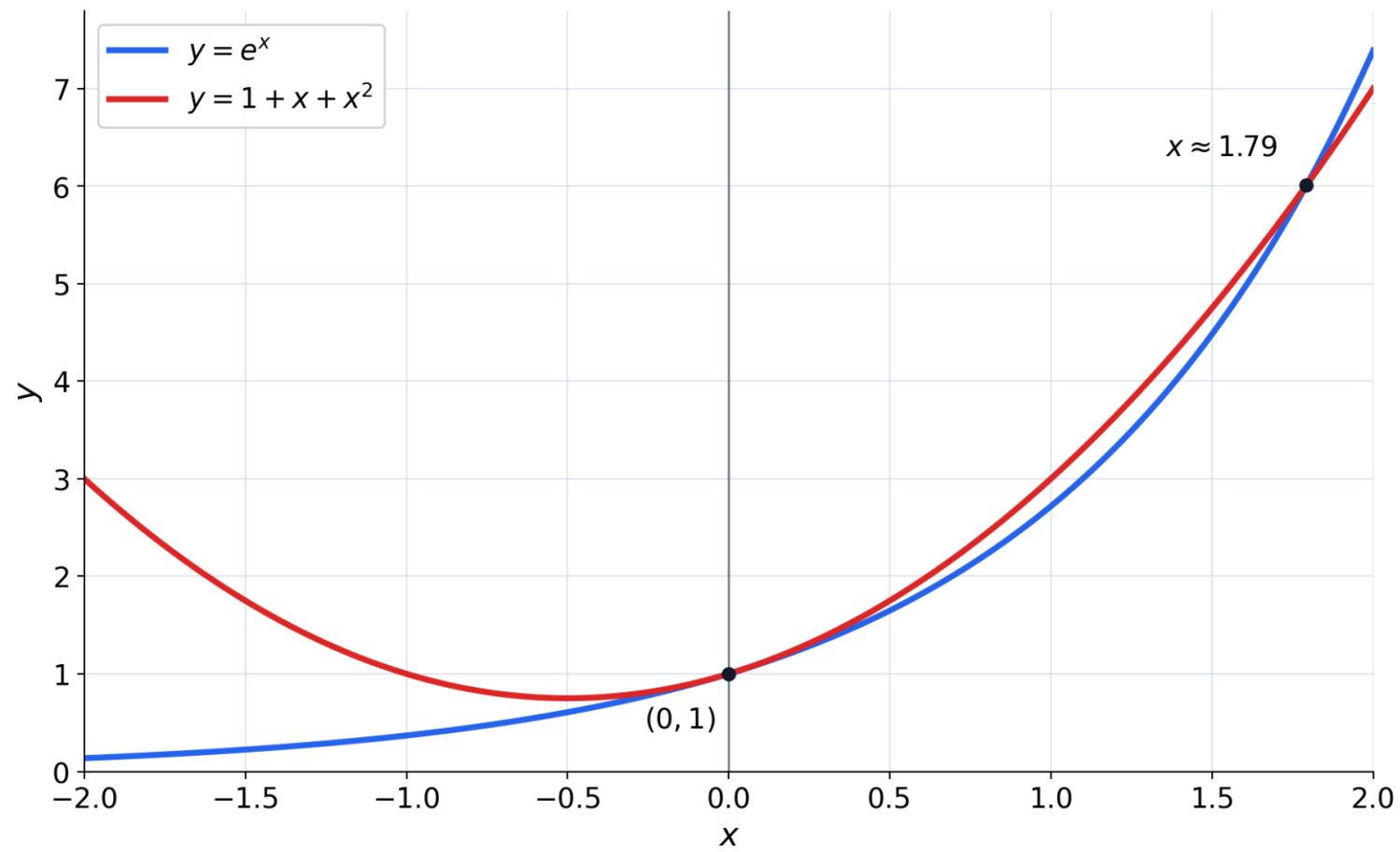
Recall $W_t = \sum_{i=1}^N \exp(-\eta L_{t-1}(i))$

$$\begin{cases} W_1 = \sum_{i=1}^N \exp(0) = N \\ W_{T+1} = \sum_{i=1}^N \exp(-\eta L_T(i)) \geq \exp(-\eta L_T(i^*)) \end{cases} \quad \underline{L_0(i) = 0}$$

$$\Rightarrow \ln \frac{W_{T+1}}{W_1} = \ln W_{T+1} - \ln W_1 \geq -\eta L_T(i^*) - \ln N$$

$$\Rightarrow -\eta L_T(i^*) - \ln N \leq -\eta \sum_{t=1}^T \langle p_t, l_t \rangle + \eta^2 \sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$

$$\Rightarrow \sum_{t=1}^T \langle p_t, l_t \rangle - \underbrace{L_T(i^*)}_{\text{min}} \leq \frac{\ln N}{\eta} + \eta \sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$



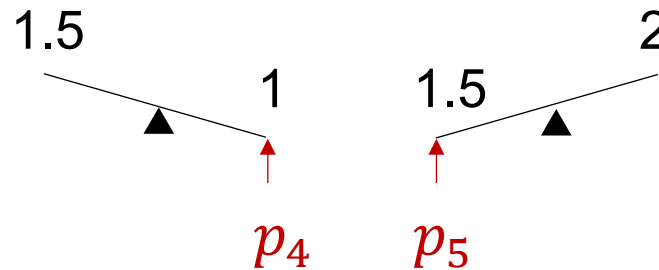
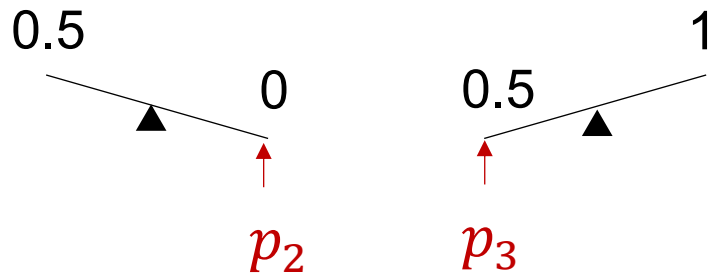
Why softmin works, while min does not?

(Discussion)

The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

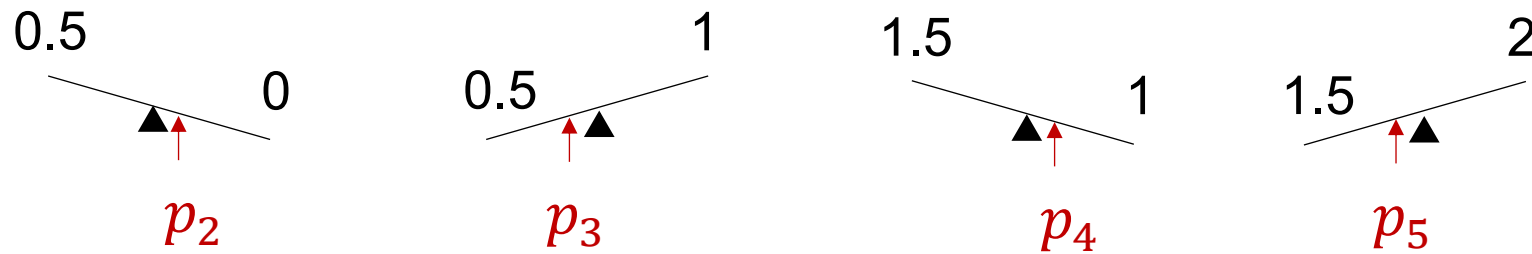
FTL: $i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$



The Failure of Follow-The-Leader

Time	$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$	$t = 6$	...
$\ell_t(1)$	0.5	0	1	0	1	0	...
$\ell_t(2)$	0	1	0	1	0	1	...

EW: $p_t(i) \propto e^{-\eta L_{t-1}(i)}$



Why Stability Helps?

(Discussion)

$$\text{dist}_{\text{ave}}(P_t, P_{t+1})$$

$\nearrow$
 l_t

Be-The-Leader Lemma

Theorem 2. Follow the leader (FTL) strategy

$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \sum_{s=1}^{t-1} f_s(x)$$

ensures

$$\sum_{t=1}^T f_t(x_t) - \min_x \sum_{t=1}^T f_t(x) = \text{Regret} \leq \sum_{t=1}^T (f_t(x_t) - f_t(x_{t+1}))$$

If we have stability:

$$x_t \approx x_{t+1}$$

(This lemma doesn't need convexity of f_t or $\mathcal{X}$)

$$\sum_{t=1}^T f_t(x_{t+1}) - \min_x \sum_{t=1}^T f_t(x) \leq 0$$

Proof of Theorem 2

$$\text{By FTL: } \chi_{t+1} = \arg \min_x \sum_{s=1}^t f_s(x)$$

Define $\phi_t = \min_x \sum_{s=1}^t f_s(x) = \sum_{s=1}^t f_s(\chi_{t+1})$

$$\phi_{t-1} = \min_x \sum_{s=1}^{t-1} f_s(x) = \sum_{s=1}^{t-1} f_s(\chi_t)$$

$$\leq \sum_{s=1}^{t-1} f_s(\chi_{t+1})$$

($\because \chi_t$ is the minimizer)

$$= \sum_{s=1}^t f_s(\chi_{t+1}) - f_t(\chi_{t+1})$$

$$= \phi_t - f_t(\chi_{t+1})$$

$$\phi_T = \min_x \sum_{t=1}^T f_t(x)$$

$$\phi_0 = 0$$

$$f_t(\chi_{t+1}) \leq \phi_t - \phi_{t-1}$$

$$\sum_{t=1}^T f_t(\chi_{t+1}) \leq \sum_{t=1}^T (\phi_t - \phi_{t-1})$$

$$\leq \sum_{t=1}^T (\phi_t - \phi_{t-1})$$

$$= \phi_T - \phi_0$$

$$\sum_{t=1}^T f_t(\chi_{t+1}) \leq \min_x \sum_{t=1}^T f_t(x)$$

$$= (\phi_1 - \phi_0) + (\phi_2 - \phi_1) + \dots + (\phi_T - \phi_{T-1})$$

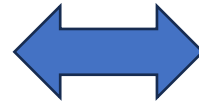
Follow The Regularized Leader

(Generalizing the Exponential Weights algorithm)

An Equivalent Form of EW

$$\psi(p) := \sum_{i=1}^N p(i) \ln p(i) \quad (\text{convex})$$

$$p_t(i) = \frac{e^{-\eta L_{t-1}(i)}}{\sum_{j=1}^N e^{-\eta L_{t-1}(j)}}$$

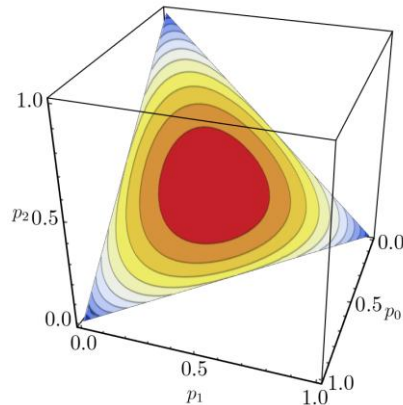


Verify yourself
(Lagrange multiplier)

$$p_t = \operatorname{argmin}_{p \in \Delta_N} \left\{ \langle p, L_{t-1} \rangle + \frac{1}{\eta} \psi(p) \right\}$$

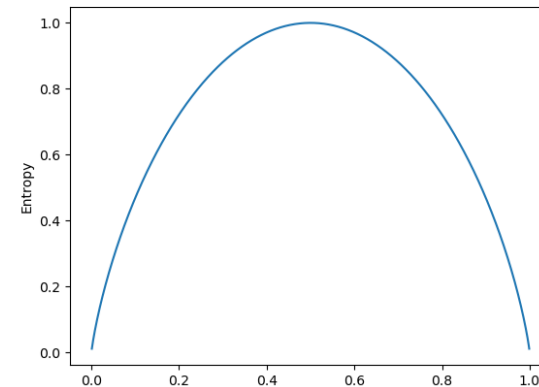
negative entropy

$-\psi(p) = \sum_{i=1}^N p(i) \ln \frac{1}{p(i)}$ is called the **(Shannon) entropy function** (concave)



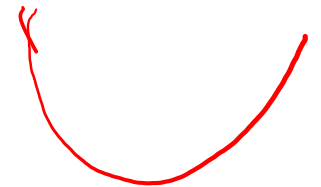
N=3

[\(link\)](#)



N=2

[\(link\)](#)



Follow the Regularized Leader (FTRL) for OCO

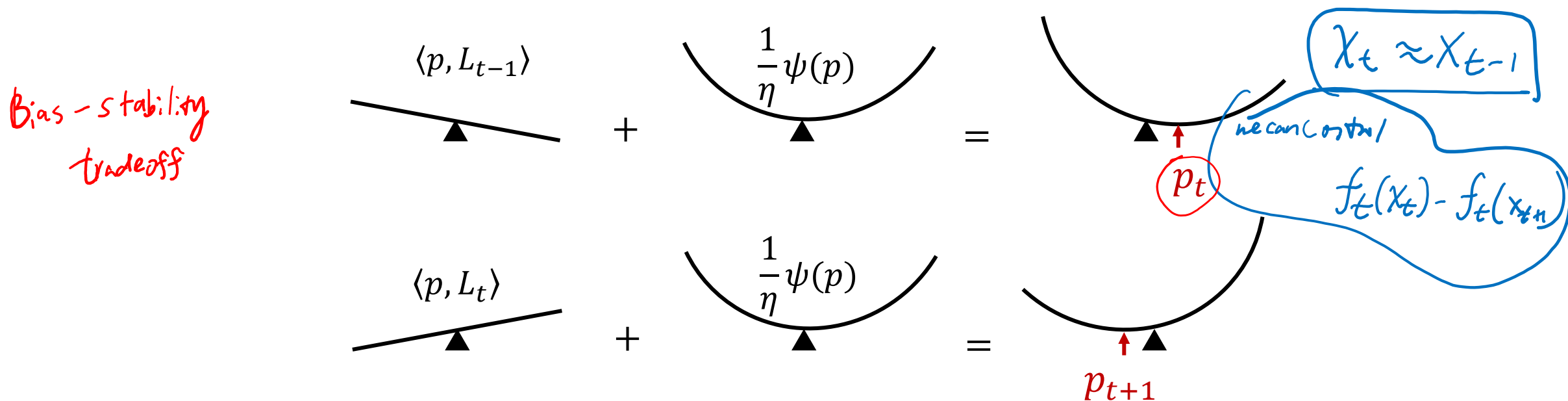
$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \sum_{s=1}^{t-1} f_s(x) + \frac{1}{\eta} \psi(x) \right\}$$

Regularizer ψ

$$\operatorname{argmin}_{x \in \mathcal{X}} \sum_{s=0}^{t-1} f_s(x)$$

define $f_0(x) = \frac{1}{\eta} \psi(x)$

If ψ is *curved*, it makes the update more stable:



Strongly Convex Functions

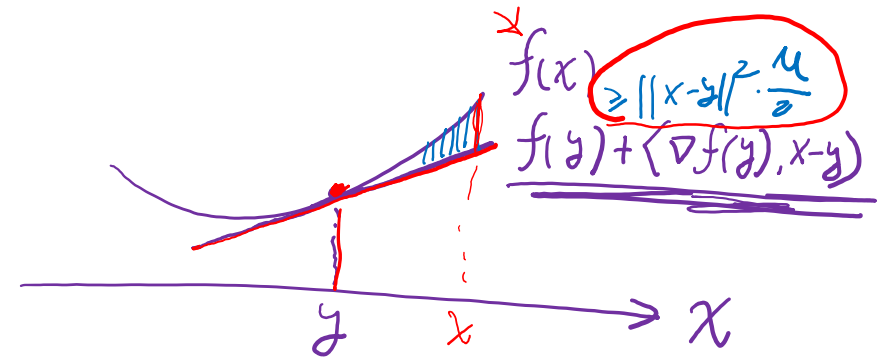
Def (convex function):

A differentiable function $\psi: X \rightarrow \mathbb{R}$ is convex if $\forall x, y \in X$

$$\psi(x) \geq \psi(y) + \langle \nabla \psi(y), x - y \rangle$$

Def (μ -strongly convex fun) ψ is called μ -strongly-convex with respect to norm $\|\cdot\|$ if $\forall x, y$

$$\underline{\psi(x)} \geq \psi(y) + \langle \nabla \psi(y), x - y \rangle + \frac{\mu}{2} \|x - y\|^2$$



Regret Guarantee of FTRL in OLO

$$\begin{aligned}\|x\|_2 &= \sqrt{x_1^2 + \dots + x_d^2} \\ \|x\|_\infty &= \max_i |x_i| \\ \|x\|_1 &= |x_1| + |x_2| + \dots + |x_d|\end{aligned}$$

Theorem 3. Let ψ be μ -strongly-convex in norm $\|\cdot\|$.

Then FTRL in OCO ensures,

$$\sum_{t=1}^T f_t(x_t) - \min_{x \in \mathcal{X}} \sum_{t=1}^T f_t(x) \leq \frac{\max_{u \in \mathcal{X}} \psi(u) - \min_{u \in \mathcal{X}} \psi(u)}{\eta} + \frac{\eta}{\mu} \sum_{t=1}^T \|\nabla f_t(x_t)\|_*^2$$

$\|\cdot\|_*$ is the dual norm of $\|\cdot\|$

$$\begin{aligned}\frac{O(1)}{\eta} + O(2) \sum_{t=1}^T G^2 &= O\left(\frac{1}{\eta} + 2TG^2\right) \\ \eta &\approx \frac{1}{G\sqrt{T}} \quad O(G\sqrt{T})\end{aligned}$$

Zinkevich. Online Convex Programming and Generalized Infinitesimal Gradient Ascent. 2003.

Zinkevich. Online Convex Programming and Generalized Infinitesimal Gradient Ascent. 2003.

Proof of Theorem 3 (Stability)

By BTL lemma,

$$\forall x, \quad \sum_{t=0}^T f_t(x_t) - \sum_{t=0}^T f_t(x) \leq \sum_{t=0}^T (f_t(x_t) - f_t(x_{t+1}))$$

$$\Rightarrow \sum_{t=0}^T f_t(x_{t+1}) \leq \sum_{t=0}^T f_t(x)$$

$$\Rightarrow \frac{1}{\eta} \psi(x_1) + \sum_{t=1}^T f_t(x_{t+1}) \leq \frac{1}{\eta} \psi(x) + \sum_{t=1}^T f_t(x)$$

$$\Rightarrow \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x) \leq \underbrace{\frac{1}{\eta} \psi(x) - \frac{1}{\eta} \psi(x_1)}_{\text{Bias } (\approx \frac{1}{\eta})} + \underbrace{\sum_{t=1}^T (f_t(x_t) - f_t(x_{t+1}))}_{\text{Stability } (\approx \eta)}$$

$$x_t = \underset{x}{\operatorname{argmin}} \left\{ \sum_{s=0}^{t-1} f_s(x) \right\}$$

$$f_0(x) = \frac{1}{\eta} \psi(x)$$

$$\langle p_t, l_t \rangle - \langle p_{t+1}, l_t \rangle$$

Given $\|\cdot\|$, $\|x\|_* \triangleq \sup_{\|y\| \leq 1} \langle x, y \rangle$

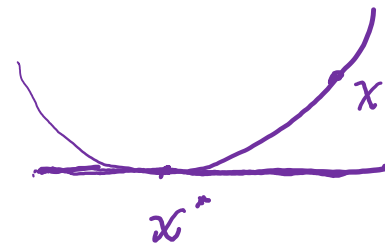
$\langle a, b \rangle \leq \|a\| \|b\|_*$

Proof of Theorem 3 (Stability)

Goal: bound $f_t(x_t) - f_t(x_{t+1}) \leq \underbrace{\langle \nabla f_t(x_t), x_t - x_{t+1} \rangle}_{\substack{\uparrow \\ \text{by convexity of } f_t}} \leq \underbrace{\|x_t - x_{t+1}\|}_{\text{red box}} \underbrace{\|\nabla f_t(x_t)\|_*}_{\text{red box}} \leq \frac{2}{\mu} \|\nabla f_t(x_t)\|_*^2$ (red circle)

Lemma: Let $F: \mathcal{X} \rightarrow \mathbb{R}$ be a μ -strongly convex function, and let $x^* = \arg\min_{x \in \mathcal{X}} F(x)$. Then for any $x \in \mathcal{X}$,

$$F(x) \geq F(x^*) + \frac{\mu}{2} \|x - x^*\|^2$$



Proof: By strong convexity, $F(x) \geq F(x^*) + \langle \nabla F(x^*), x - x^* \rangle + \frac{\mu}{2} \|x - x^*\|^2$

By first-order optimality $\langle \nabla F(x^*), x - x^* \rangle \geq 0$ for any x

Proof of Theorem 3

$$x_t = \operatorname{argmin}_{x \in X} \left\{ \sum_{s=1}^{t-1} f_s(x) + \frac{1}{\eta} \psi(x) \right\}$$

$$F_t(x)$$

F_t is $\frac{\mu}{\eta}$ -strongly convex

$$F_{t+1}(x) - F_t(x) = f_t(x)$$

By the previous lemma :

$$\begin{cases} F_t(x_{t+1}) \geq F_t(x_t) + \frac{\mu}{2\eta} \|x_t - x_{t+1}\|^2 \\ F_{t+1}(x_t) \geq F_{t+1}(x_{t+1}) + \frac{\mu}{2\eta} \|x_t - x_{t+1}\|^2 \end{cases}$$

$$F_t(x_{t+1}) - F_{t+1}(x_{t+1}) + F_{t+1}(x_t) - F_t(x_t) \geq \frac{\mu}{\eta} \|x_t - x_{t+1}\|^2$$

$$\Rightarrow \frac{\mu}{\eta} \|x_t - x_{t+1}\|^2 \leq f_t(x_t) - f_t(x_{t+1}) \leq \|x_t - x_{t+1}\| \| \nabla f_t(x_t) \|_*$$

$$\Rightarrow \|x_t - x_{t+1}\| \leq \frac{\eta}{\mu} \| \nabla f_t(x_t) \|_*$$

Special Case: Negative Shannon Entropy

$$\mathcal{X} = \Delta_N, \quad \psi(x) = \sum_{i=1}^N x(i) \log x(i)$$

(Pinsker's ineq)

ψ is 1-strongly concave w.r.t. $\|\cdot\|_1$

$$\psi(x) \in [-\log N, 0]$$

$$f_t(x) = (x, l_t)$$

$$\nabla f_t(x) = l_t$$

$$\text{Regret} \leq \frac{\ln N}{2} + 2 \sum_{t=1}^T \|l_t\|_\infty^2$$

(max_i l_t(i)²)

previous

$$\sum_{t=1}^T \sum_{i=1}^N p_t(i) l_t(i)^2$$

Special Case: Squared Euclidean Norm

$\mathcal{X}$ = general convex set, $\psi(x) = \frac{1}{2} \|x\|_2^2$

ψ is 1-strongly convex w.r.t. $\|\cdot\|_2$

Assume $\|x\|_2 \leq R \quad \forall x \in \mathcal{X}$

$$R_{\text{opt}} \leq \frac{\frac{1}{2} R^2}{\gamma} + \gamma \sum_{t=1}^T \|\nabla f_t(x_t)\|_2^2$$

$$\operatorname{argmin}_{x \in \mathcal{X}} \left\| \underbrace{(x_1 - \gamma L_{t-1})}_{\text{subproblem}} - x \right\|_2 \iff$$

$$\operatorname{argmin}_{x \in \mathcal{X}} \left\{ \langle x, L_{t-1} \rangle + \frac{1}{2} \psi(x) \right\}$$

Remark: The Role of Entropy Regularization

Exploration: $p_t(i) > 0 \quad \forall i$

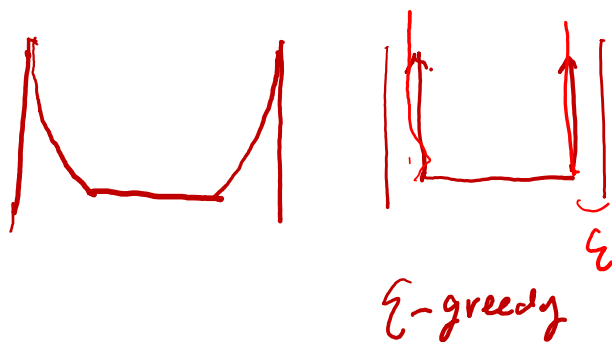
Stability: $p_t \approx p_{t+1}$

for the case when we have limited feedback

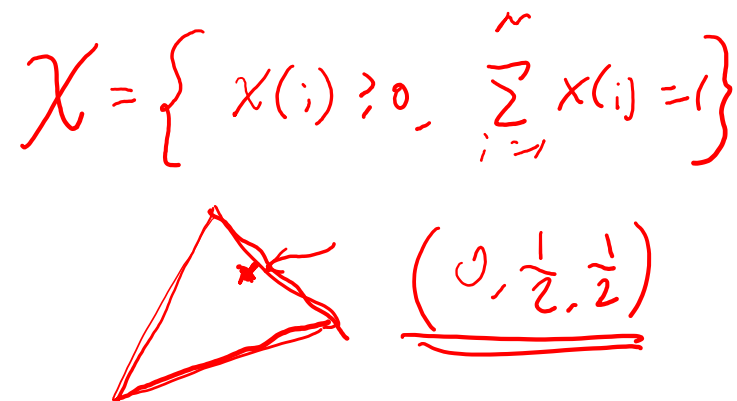
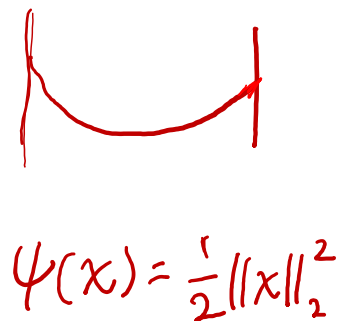
(ψ has steep boundary)

(ψ has curvature) *← property we use here*

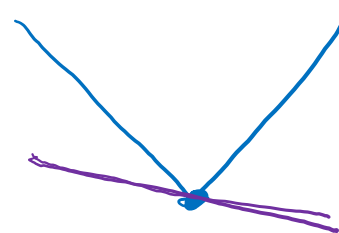
Exploratory but not stable:



Stable but not exploratory:



OCO with only gradient feedback



Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For $t = 1, 2, \dots, T$:

Learner makes a decision $x_t \in \mathcal{X}$

Adversary decides a convex loss function $f_t: \mathcal{X} \rightarrow \mathbb{R}$

Adversary reveals $\nabla f_t(x_t)$ to the learner

create $g_t(x) = \langle x, \nabla f_t(x_t) \rangle$

run FTRL on $g_t(x)$

Why this works?

$$g_t(x) = f_t(x_t) + \langle x - x_t, \nabla f_t(x_t) \rangle$$

convexity

$$g_t(x) = \langle x, \nabla f_t(x_t) \rangle$$

$$\sum_t f_t(x_t) - f_t(x^*) \leq \langle x_t - x^*, \nabla f_t(x_t) \rangle = \sum_t g_t(x_t) - g_t(x^*) \leq o(T)$$

Online Mirror Descent

Online Mirror Descent

Implicit Online Mirror Descent

$$x_{t+1} = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ f_t(x) + \frac{1}{\eta} D_\psi(x, x_t) \right\}$$

Online Mirror Descent

(Problem 3 of Homework 1)

$$x_{t+1} = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} D_\psi(x, x_t) \right\}$$

The regret guarantee and analysis are closely related to those of FTRL

Follow The Perturbed Leader

Online Linear Optimization with a Weaker Adversary

Given: convex set $\mathcal{X} \subset \mathbb{R}^d$

For $t = 1, 2, \dots, T$:

Adversary decides a **linear** loss function $f_t(x) = \langle x, \ell_t \rangle$
(not revealing it to the learner at this moment)

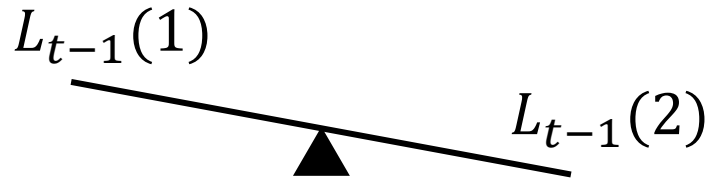
Learner makes a decision $x_t \in \mathcal{X}$

Adversary reveals ℓ_t to the learner



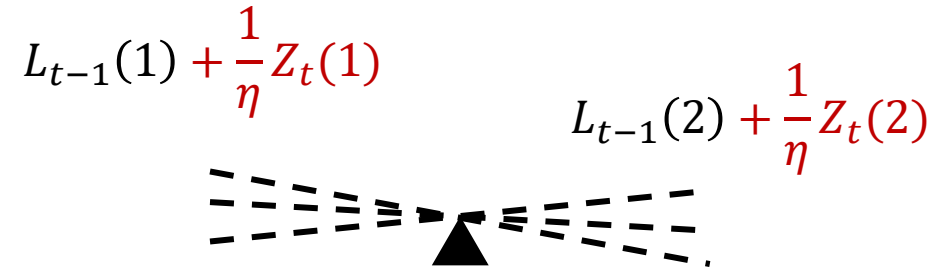
order swapped

Another Way to Avoid Exploitation by Adversary



$$i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i)$$

Follow the Leader



For example: $Z_t(i) \sim \mathcal{N}(0, \sigma^2) \quad \forall i = 1, 2$

$$i_t = \operatorname{argmin}_{i \in [N]} L_{t-1}(i) + \frac{1}{\eta} Z_t(i)$$

effective $P_t(i) = \Pr \left\{ i = \operatorname{argmin}_{i \in [N]} L_{t-1}(i) + \frac{1}{\eta} Z_t(i) \right\}$

Follow the Perturbed Leader

$\|P_t - P_{t+1}\|$

Similar to FTRL, “randomization” does not precisely capture the key mechanism of FTPL. The key is “stability.”

FTPL for OLO

Follow The Perturbed Leader (FTPL) for OLO

$$x_t = \operatorname{argmin}_{x \in \mathcal{X}} \left\{ \sum_{s=1}^{t-1} \langle x, \ell_s \rangle + \frac{1}{\eta} \langle x, Z_t \rangle \right\} = \operatorname{argmin}_{x \in \mathcal{X}} \left\langle x, L_{t-1} + \frac{1}{\eta} Z_t \right\rangle$$

where $Z_t \in \mathbb{R}^d$ is drawn from some distribution $\mathcal{D}$.

When is FTPL more preferred than FTRL?

For many common decision set $\mathcal{X}$, it is (computationally) much easier to solve

$$\operatorname{argmin}_{x \in \mathcal{X}} \langle x, L \rangle$$

than

$$\operatorname{argmin}_{x \in \mathcal{X}} \langle x, L \rangle + \psi(x)$$

for a given $L \in \mathbb{R}^d$.

When is FTPL more preferred than FTRL?

Example: Online Shortest Path

Given: a graph, a source, and a sink

Let $\mathcal{Y} \subset \{0,1\}^{|E|}$ be the set of paths from the source to sink

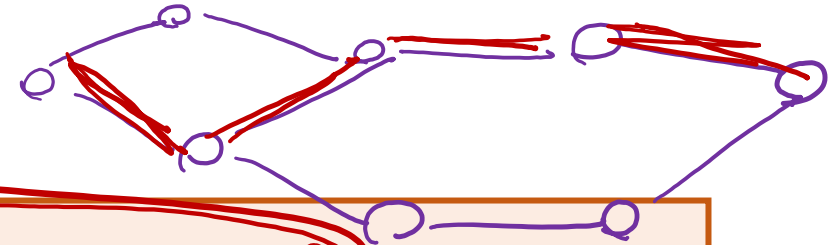
For $t = 1, 2, \dots, T$:

Adversary decides the costs on all edges $\ell_t \in \mathbb{R}^{|E|}$

Learner chooses a path $y_t \in \mathcal{Y}$

Adversary reveals ℓ_t to the learner

$|E| = 9$



$$y_t = \{1, 0, 1, 1, 0, 0, 0, 1, 0\}$$

$\mathcal{Y}$ above is not a convex set, but we can

- Run OCO in $\mathcal{X} = \text{convhull}(\mathcal{Y})$ (becoming a set of flows)
- Sample y_t such that $\mathbb{E}[y_t] = x_t$ in round t .

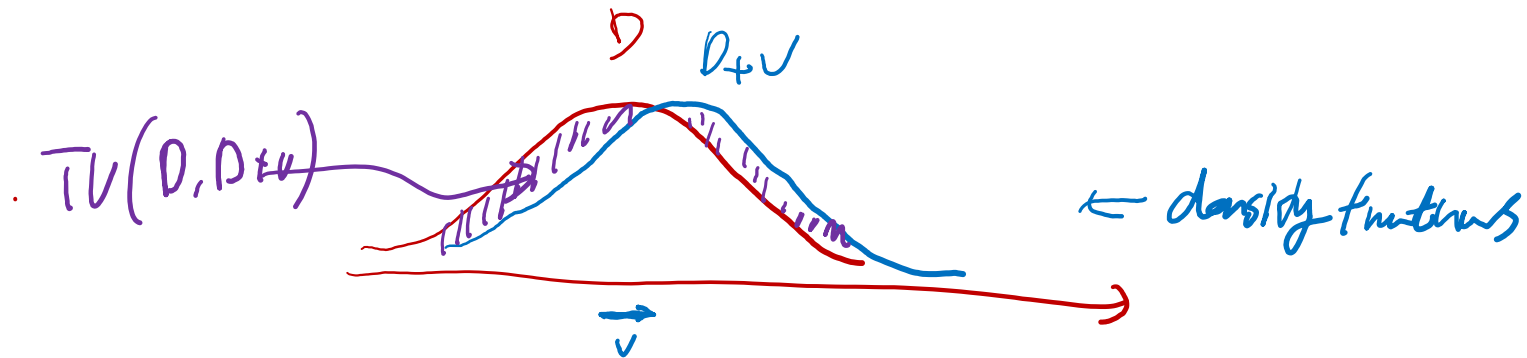
“Smooth” Perturbations

Let D be a distribution on $\mathbb{R}^d$ (we draw $Z \sim D$)

We say D is K -smooth if for any $V \in \mathbb{R}^d$

$$\underline{TV(D, D+V)} \leq \underline{K \|V\|_*}$$

where $D+V$ is a distribution such that $Z \sim D+V \equiv Z' \sim D$ and set $Z = Z' + V$



Regret Guarantee of FTPL in OLO

Theorem 4. Let $\mathcal{D}$ be a κ -smooth distribution.

Then FTPL in OLO ensures,

$$\sum_{t=1}^T \langle x_t, \ell_t \rangle - \min_{x \in \mathcal{X}} \sum_{t=1}^T \langle x_t, \ell_t \rangle \leq \frac{\mathbb{E}_{Z \sim \mathcal{D}} \left[\max_{u \in \mathcal{X}} \langle u, Z \rangle \right] - \mathbb{E}_{Z \sim \mathcal{D}} \left[\min_{u \in \mathcal{X}} \langle u, Z \rangle \right]}{\eta} + \eta \kappa \sum_{t=1}^T \|\ell_t\|_*^2$$

(Problem 4 of HW1)

Recap

- Formulation of online convex optimization (OCO) and its special cases: expert problem, online linear optimization (OLO)
- Algorithms with no-regret guarantees
 - Exponential weights
 - Follow the regularized leader
 - Follow the perturbed leader
- Key algorithmic elements: stability and bias trade-off

Know more about the history?

<https://parameterfree.com/lecture-notes-on-online-learning/>